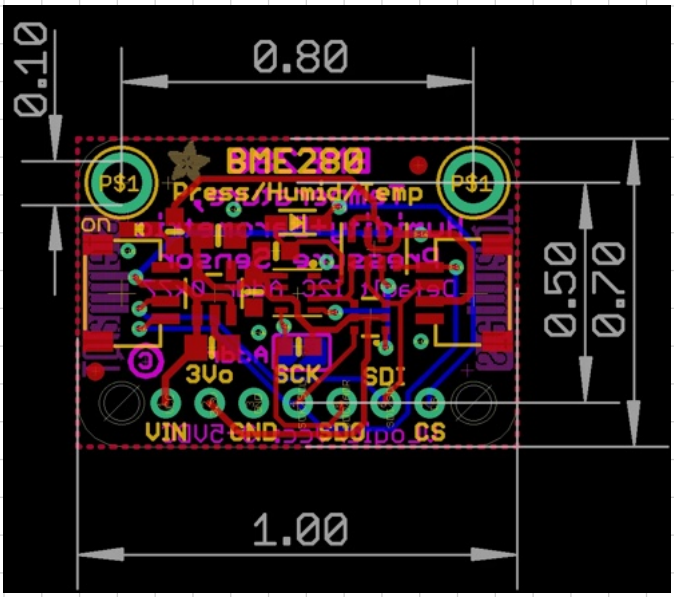
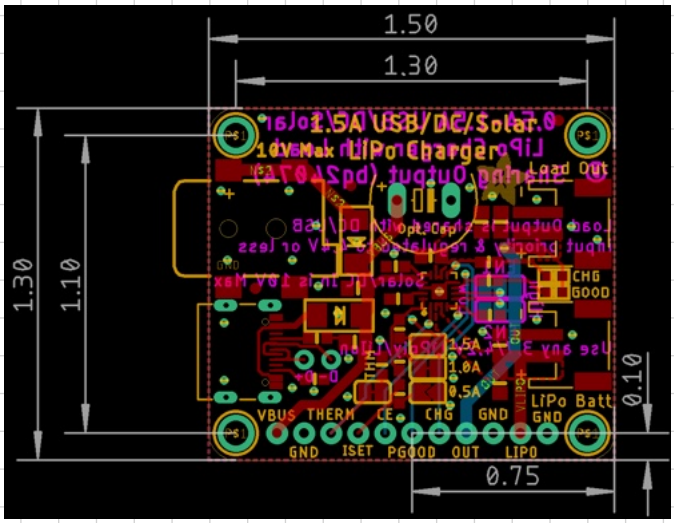
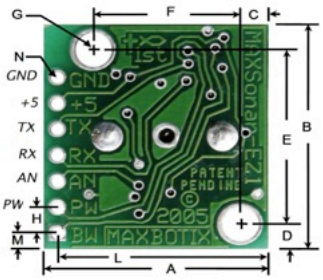
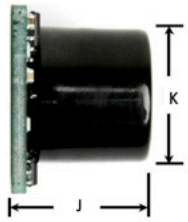




LV-MaxSonar-EZ Mechanical Dimensions



A	0.785"	19.9 mm	H	0.100"	2.54 mm
B	0.870"	22.1 mm	J	0.610"	15.5 mm
C	0.100"	2.54 mm	K	0.645"	16.4 mm
D	0.100"	2.54 mm	L	0.735"	18.7 mm
E	0.670"	17.0 mm	M	0.065"	1.7 mm
F	0.510"	12.6 mm	N	0.038" dia.	1.0 mm dia.
G	0.124" dia.	3.1 mm dia.	weight, 4.3 grams		

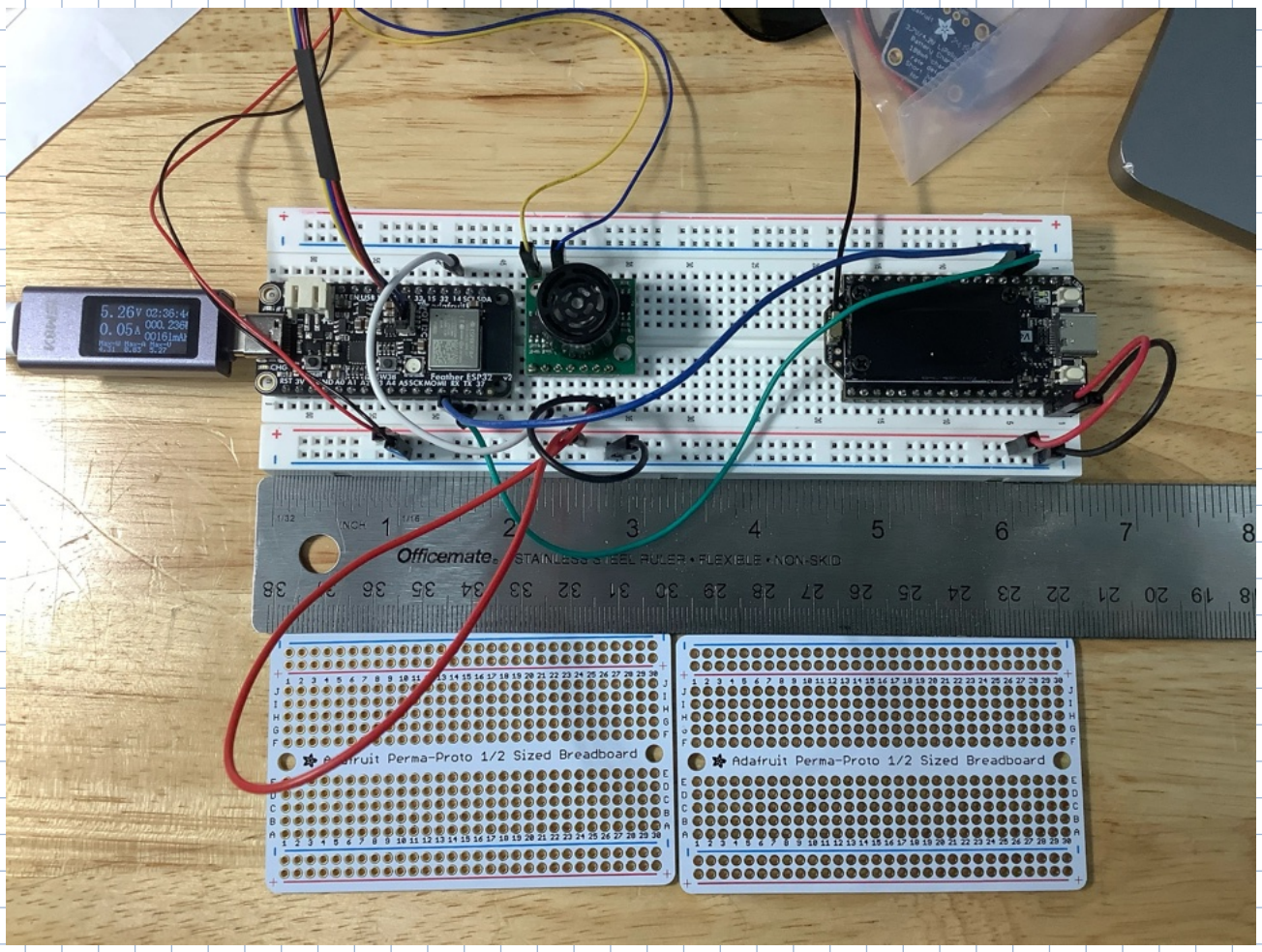


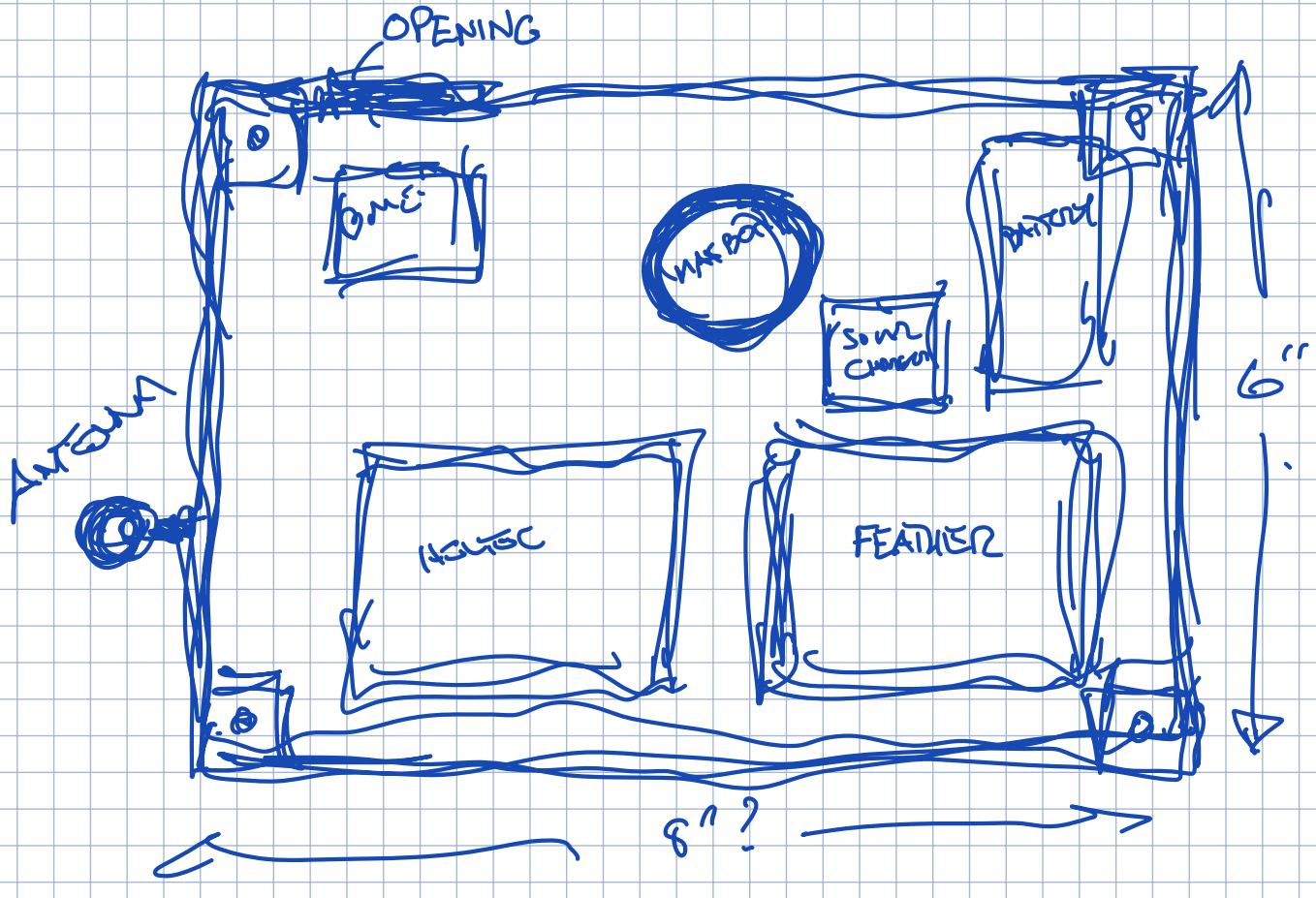
PROTO BOARDS: 2" x 3.2"

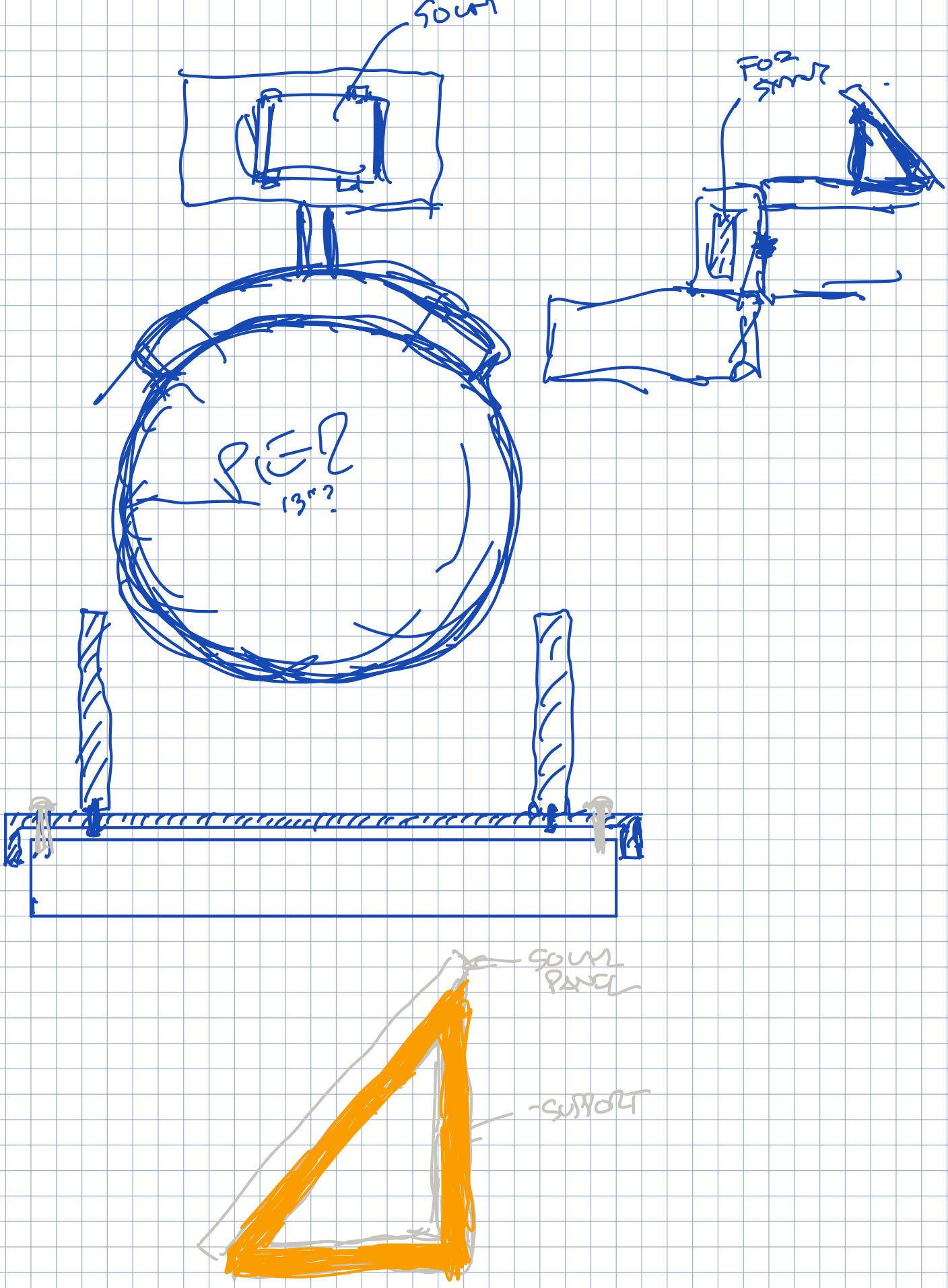
MOUNTING HOLES 2.9"

30 ROWS

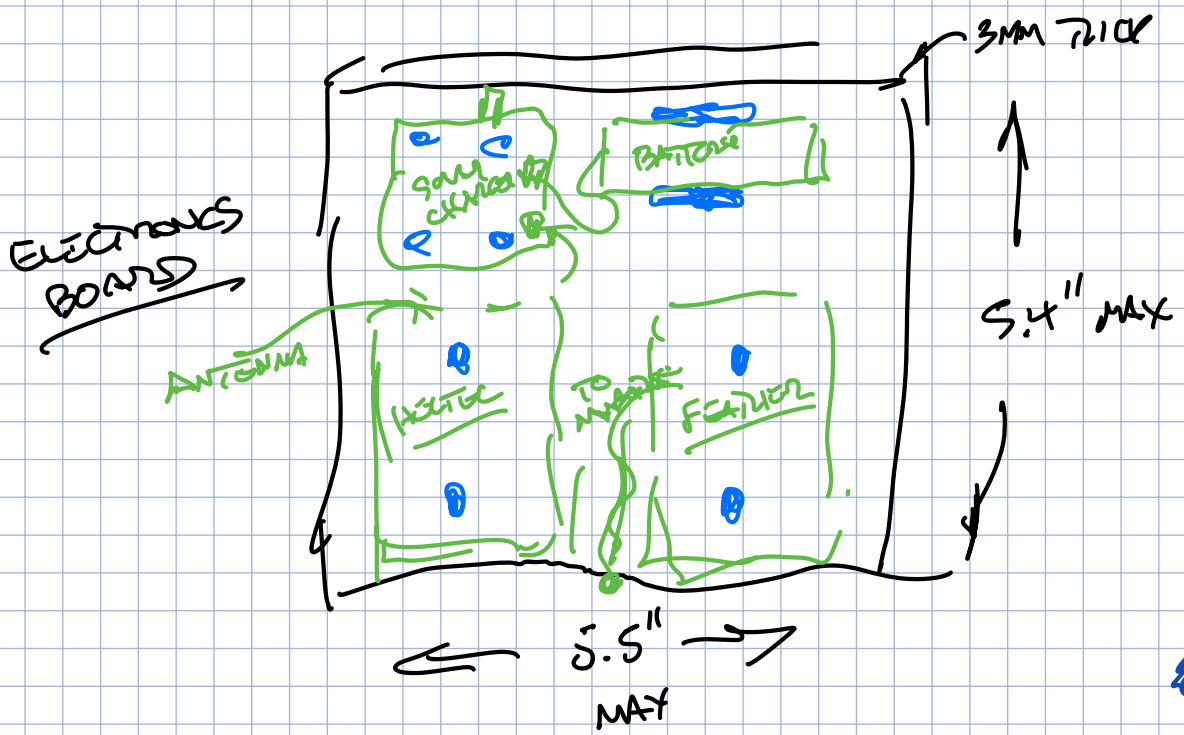
1.2MM DIA HOLES

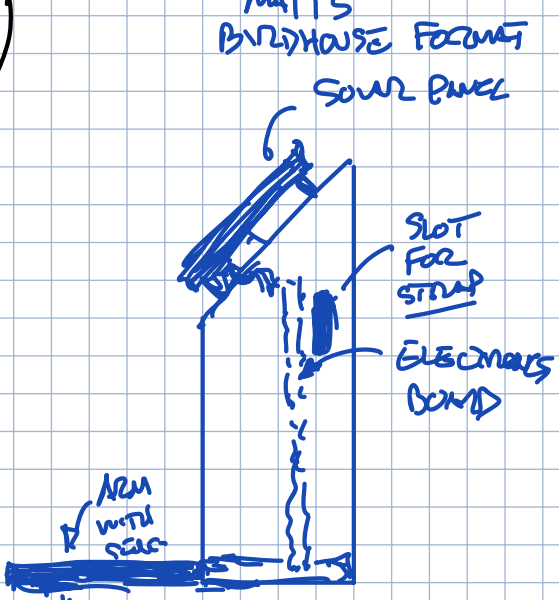

Officemate STAINLESS STEEL RULER • FLEXIBLE • NON-SKID
Adafruit Perma-Proto 1/2 Sized Breadboard
Adafruit Perma-Proto 1/2 Sized Breadboard


OPENING
INMP441
FEATHER
6"
8"?

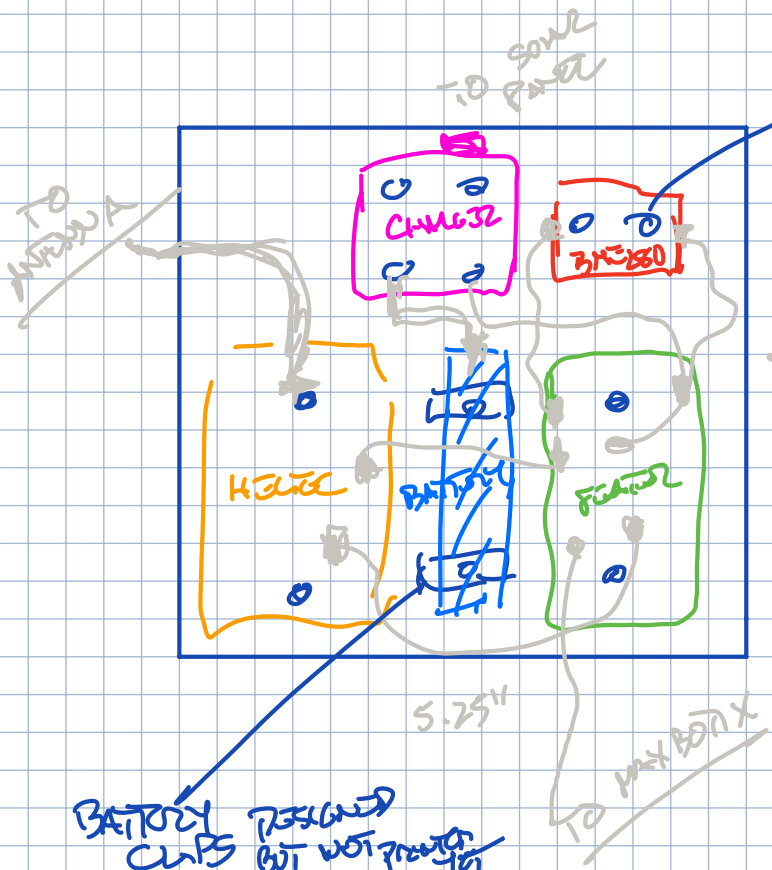

SOUND
FOR
SUPPORT
PIER
13' ?
SOUND
PANEL
-SUPPORT

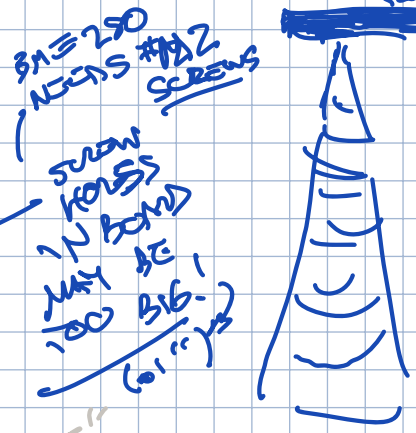
MATT WANTS TO BUILD A "BIRDHOUSE" CONTAINER FOR SENSOR, SO I WILL BUILD A FLAT BOARD FOR ELECTRONICS INSTEAD & INSERT IT INTO THE BOX





HOW DO WE CONNECT & PLACE BME 280 AND LORA ANTENNA? (AND MAXBOTIX SENSOR?)



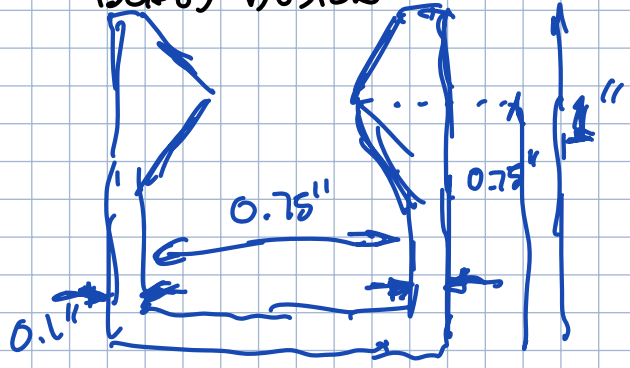


9/21/26

MADE BOARD IN FUSION & PRINTED IT WITH BAMBU

NEXT VERSION OF BOARD:

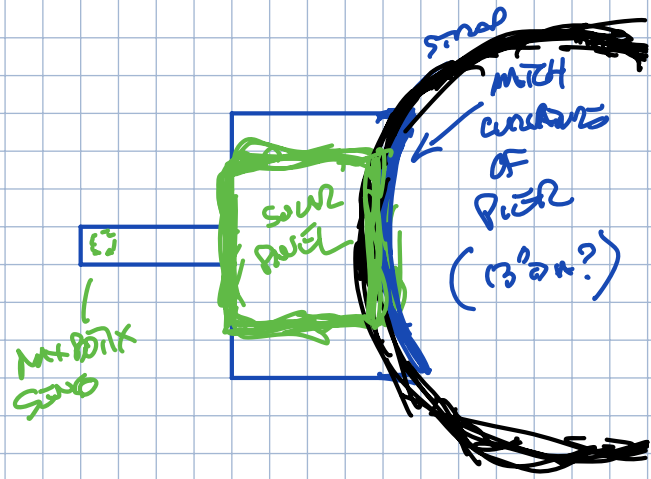
- SMALLER HOLES FOR BME280 AND SOLAR CHARGER
- 5.4" x 5.4" → LARGER TO CENTER EXPAND BOARD THINGS
- INTEGRATE BATTERY CLIPS INTO BOARD DESIGN



CAN I DESIGN CLIPS ON DIFFERENT AXES AND THEN MOVE THEM INTO PLACE FOR A SINGLE 3D PRINT?

YES!



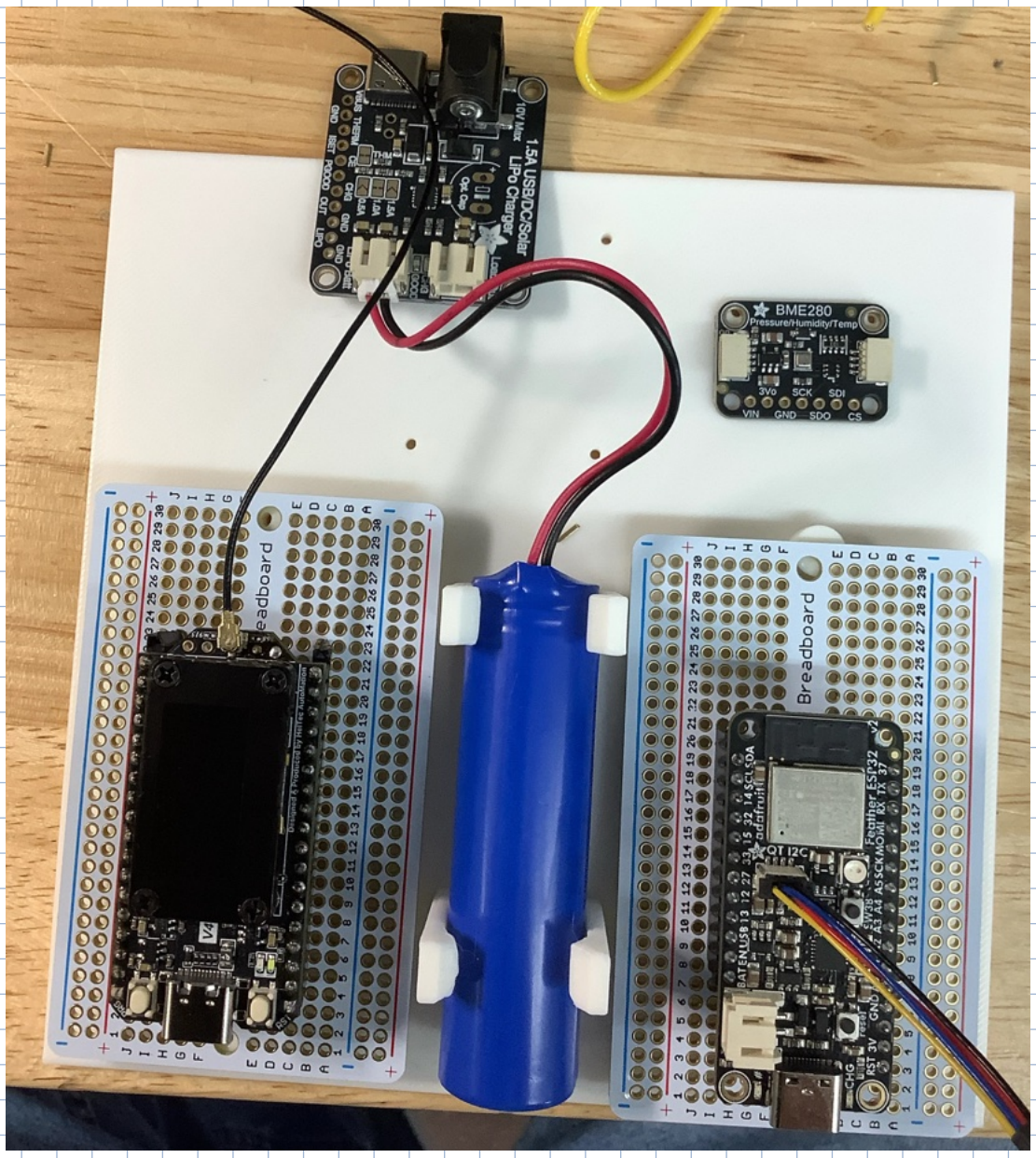


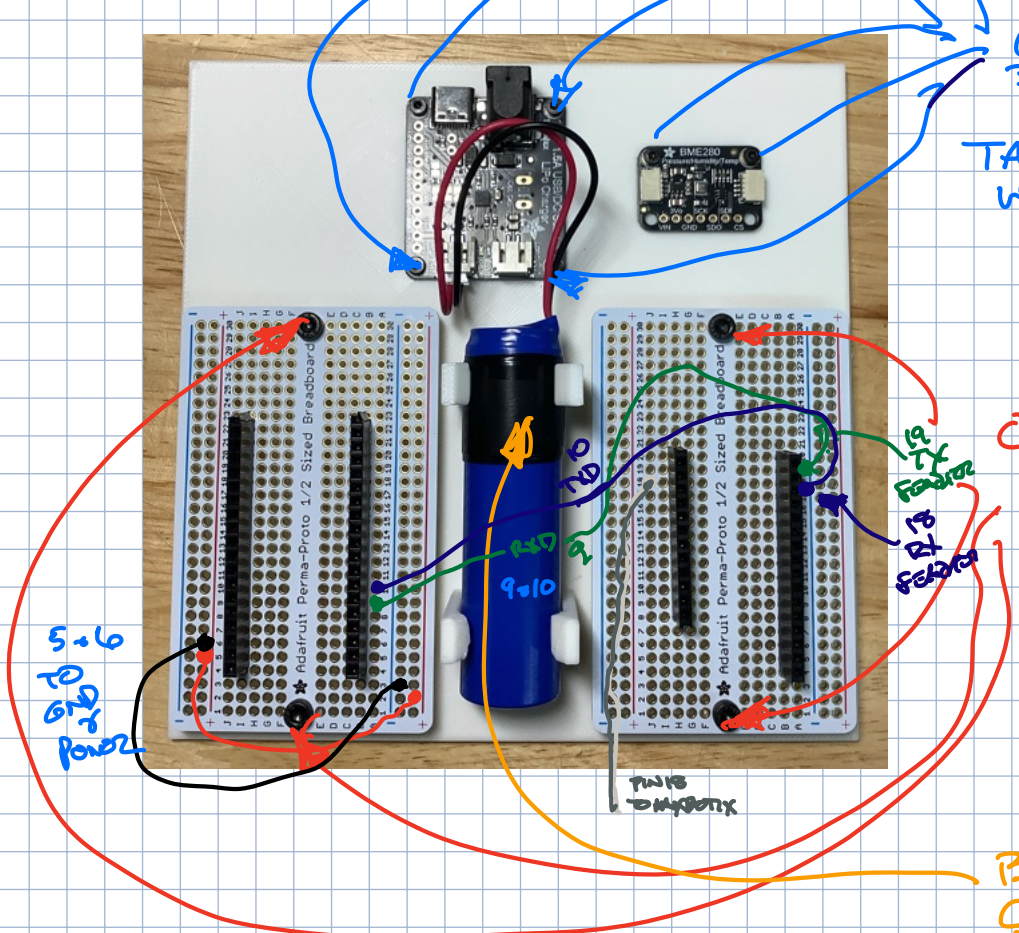
9/23/20

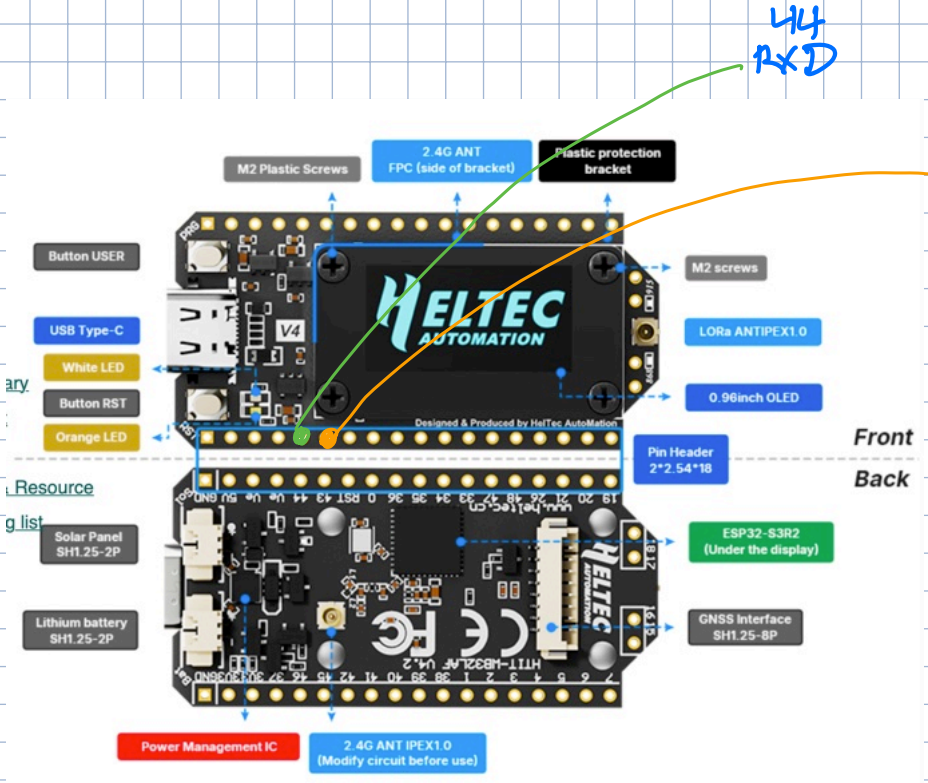
PRINTING NEW BOARD!

- 5.4 x 5.4 in
- .08" HOLES FOR BME280 AND SOLAR CHARGER
- MOVED HOLES FOR BOARDS
- BATTERY CLIPS INTEGRAL TO BOARD

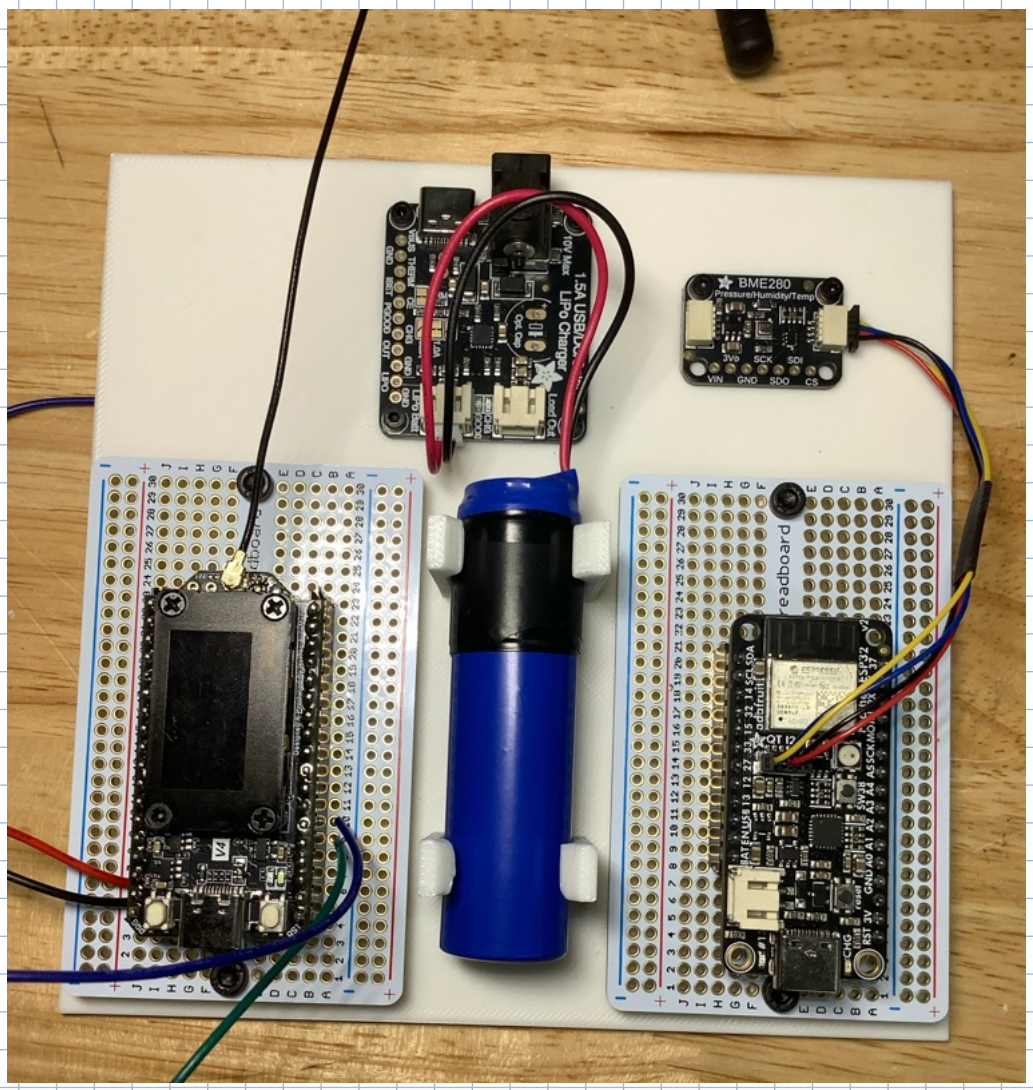
NOW THINK ABOUT WIRING




0.08" PRINTED HOLES, TAPPED WITH M2
0.10" PRINTED HOLES M3 TAPPED
5+6 TO GND & POWER
9+10
BATTERY CLIPS ARE A BIT LOOSE, SO A FEW WRAPS OF ELECTRICAL TAPE ARE USED TO MAKE SNUG FIT


44 RXD
43 TXD
M2 Plastic Screws
2.4G ANT FPC (side of bracket)
Plastic protection bracket
Button USER
USB Type-C
White LED
Button RST
Orange LED
M2 screws
LORa ANTIPEX1.0
0.96inch OLED
Pin Header 2*2.54*18
Front
Back
ESP32-S3R2 (Under the display)
GNSS Interface SH1.25-8P
Solar Panel SH1.25-2P
Lithium battery SH1.25-2P
Power Management IC
2.4G ANT IPEX1.0 (Modify circuit before use)

9/27/24



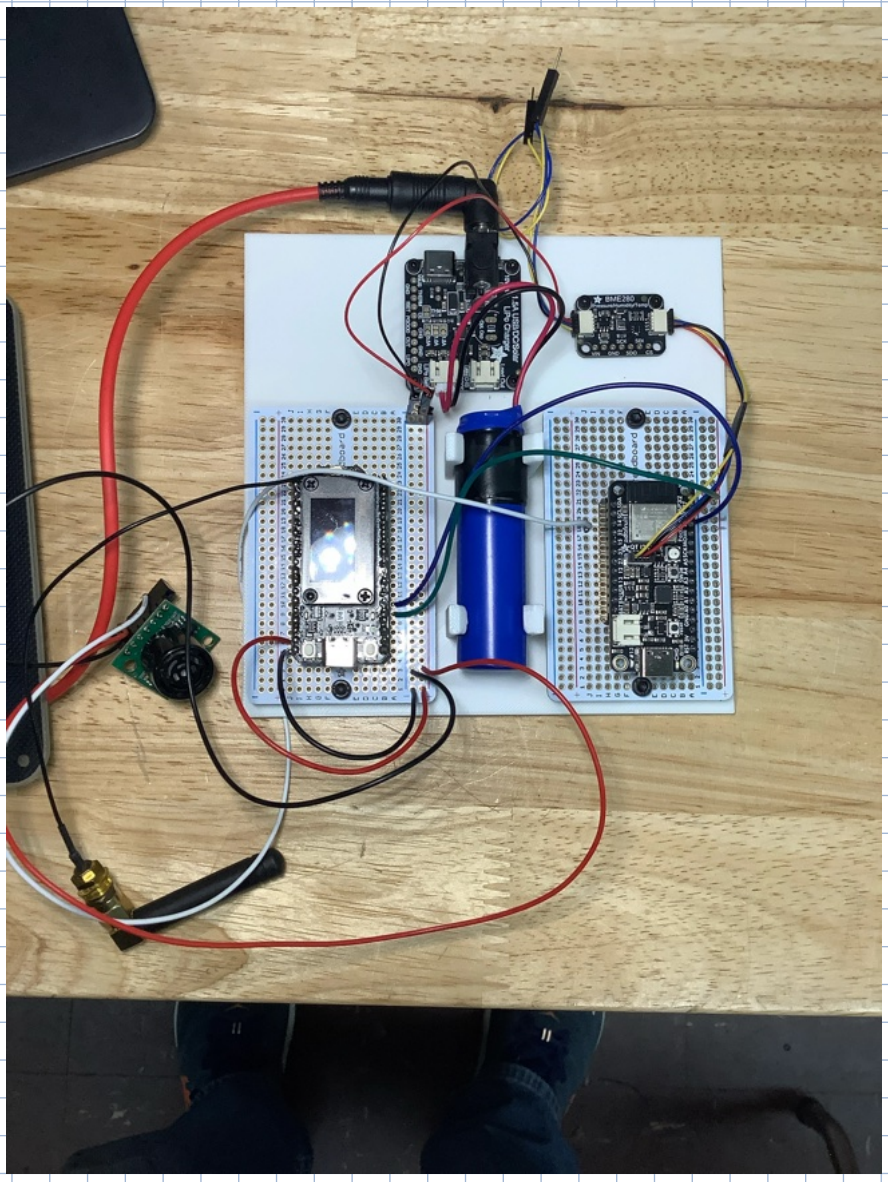
① SOLDERED HEADERS FOR HELTEC AND FEATHER TO BOARDS

② SOLDERED POWER & SERIAL DATA WIRES TO HELTEC BOARD

LOOKS GOOD SO FAR!

WIRES SO FAR:

- BLUE IS HELTEC TXD (TO FEATHER RX)
- GREEN IS HELTEC RXD (TO FEATHER TX) (SERIAL1)
- RED IS POWER (CONTROLLED THROUGH I²C)
- BLACK IS GROUND



9/28/26

COMPLETED WIRING

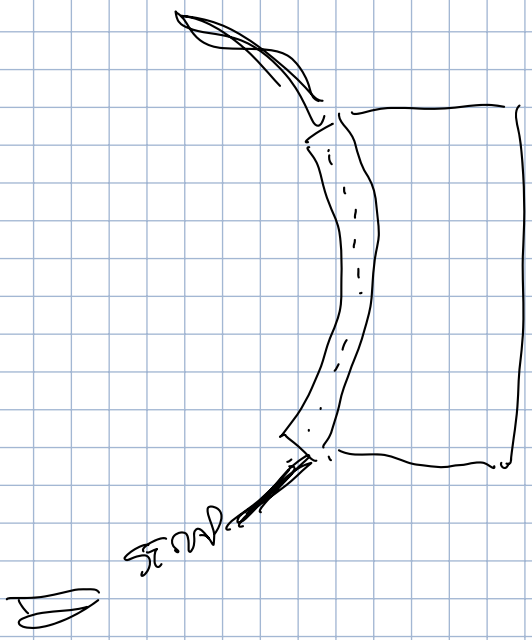
ALL EXCEPT JST CONNECTORS FOR CHARGER TO FEATHER (ON ORDER)

TESTED WITH USB C POWER FROM LAPTOP AND TIDSS[illegible]25 TEST PROGRAM

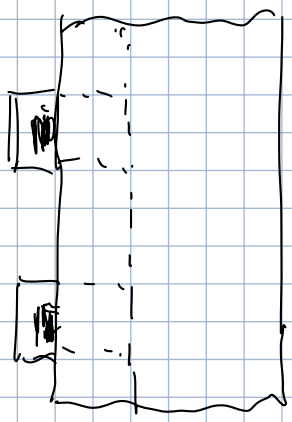
WORKS!

- NEXT STEP. TEST NEWEST SOFTWARE
- FINISH BOX AND TEST WITH WATER
- THEN INSTALL!

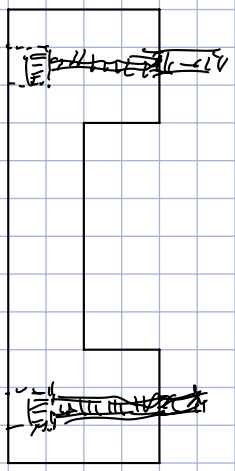
- MEASURE PIER DIAMETER CAREFULLY
- FIGURE OUT HOW TO CUT AT AN ANGLE FOR ROOF
- DESIGN STRAP HOLDERS
- DESIGN ROOF + SOLAR PANEL BRACKET
- BEEF UP WALL THICKNESS ON BOX TO SUPPORT STRAP HOLDER

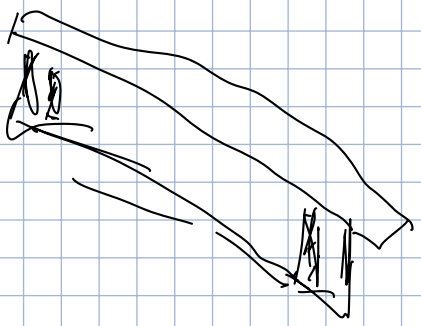

STAND

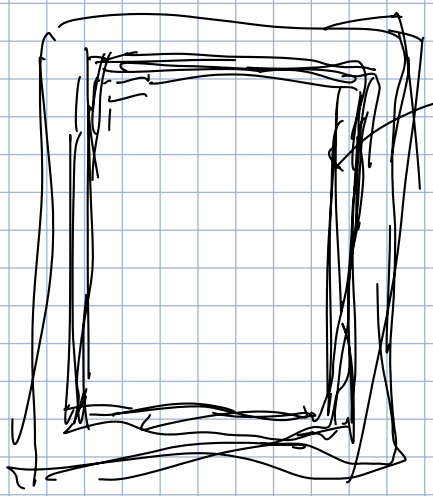











SLOT
TO FIT
WALLS



MAXBOTIX SENSOR IN EXTENSION BOX

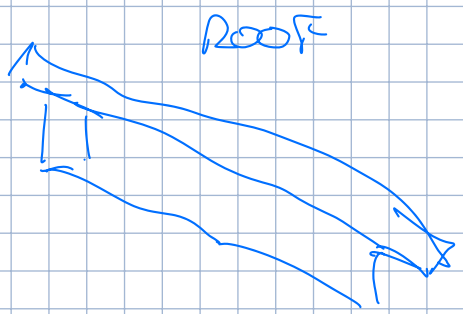
62" DISTANCE TO FLOOR

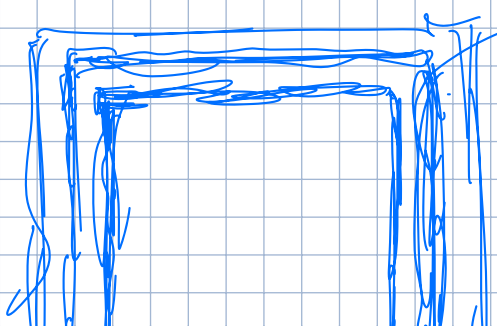
READS 155 cm (61") OK!

BOUGHT WRONG JST (BATTERY) CONNECTORS!

I NEED JST-PH

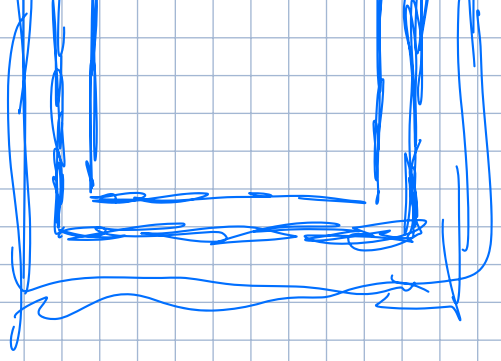
-NOT JST-XH





BOX WALLS SLIP IN BETWEEN EDGE 2 SHORT WALLS

UPSIDE-DOWN VIEW OF ROOF



10/4/26

MODIFICATIONS / IMPROVEMENTS

1) HOLES TO HOLD EXTENSION BOX TOGETHER
   - USE M4 SCREWS -
   - TAP HOLE CHANGED FROM 0.10 TO 0.12

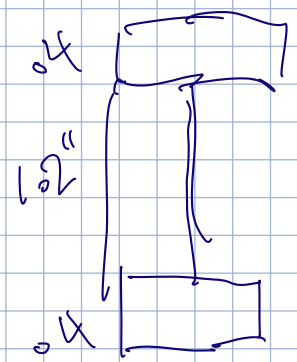
2) MEASURED PIERS PRECISELY
   ANG. 46" CIRCUMFERENCE
   14.7"

14.7 INNER DIAMETER OF SNAP BRACKET

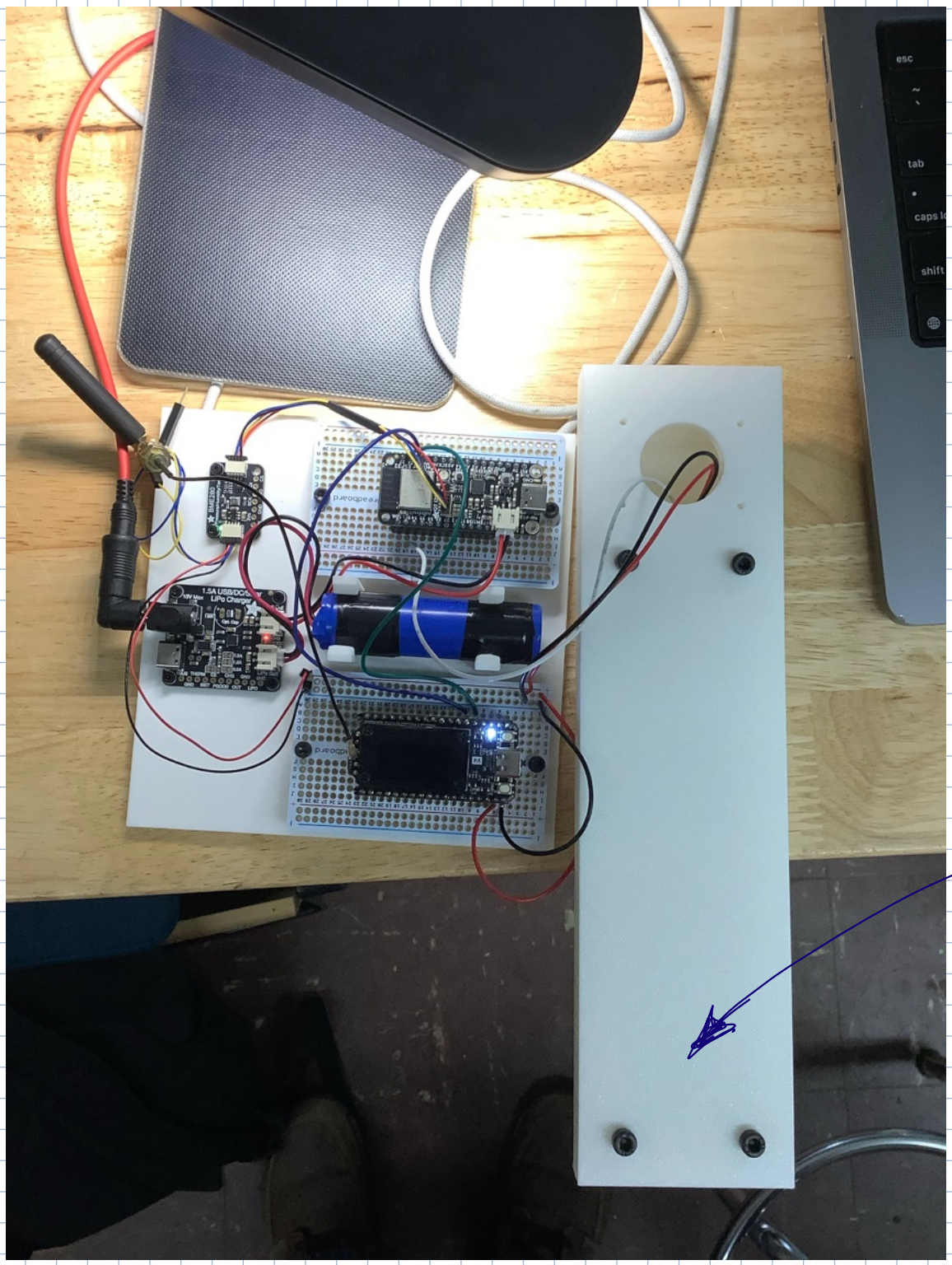
14.7 + 0.5 = 15.2" DIAMETER OF BIRDHOUSE BOX

3) ADDED CIRCLES TO CREATE ROOF AND SNAP BRACKET

4) ADDED SKETCH TO MAKE SOLID TO SUBTRACT SLANTED PART OF WALLS



10/14/26



COMPLETED BOARD

WITH SOLAR POWER

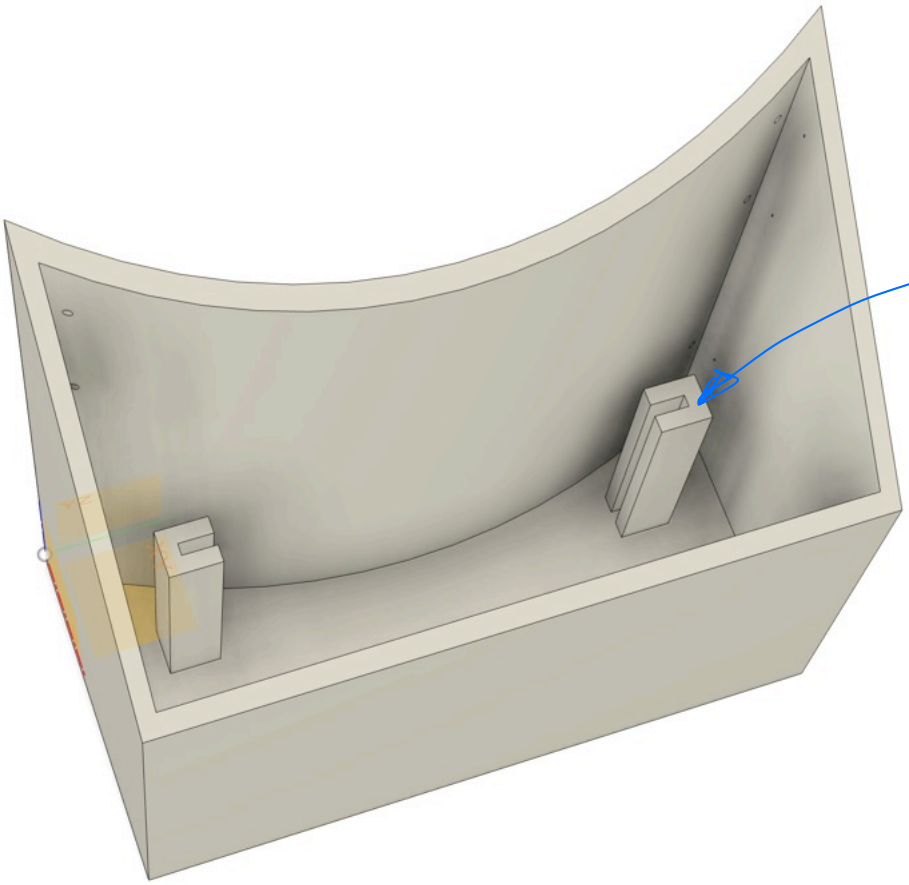
USING TEST SOFTWARE

SENSOR IS HERE

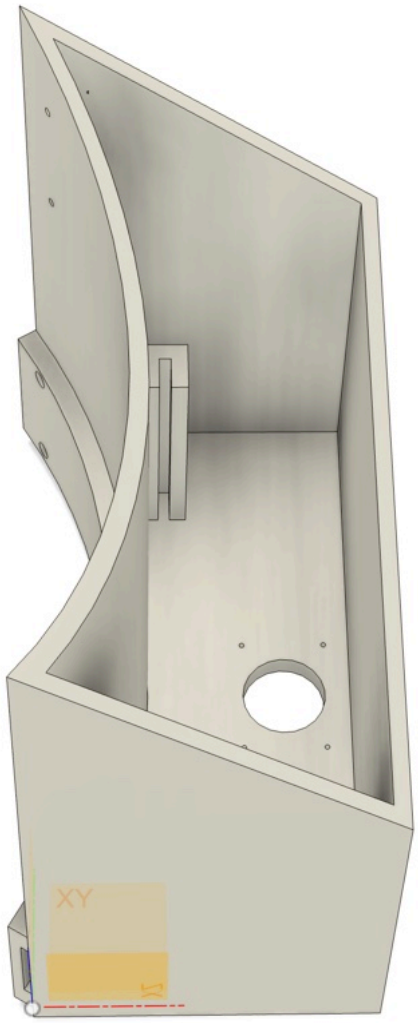
37" FROM FLOOR

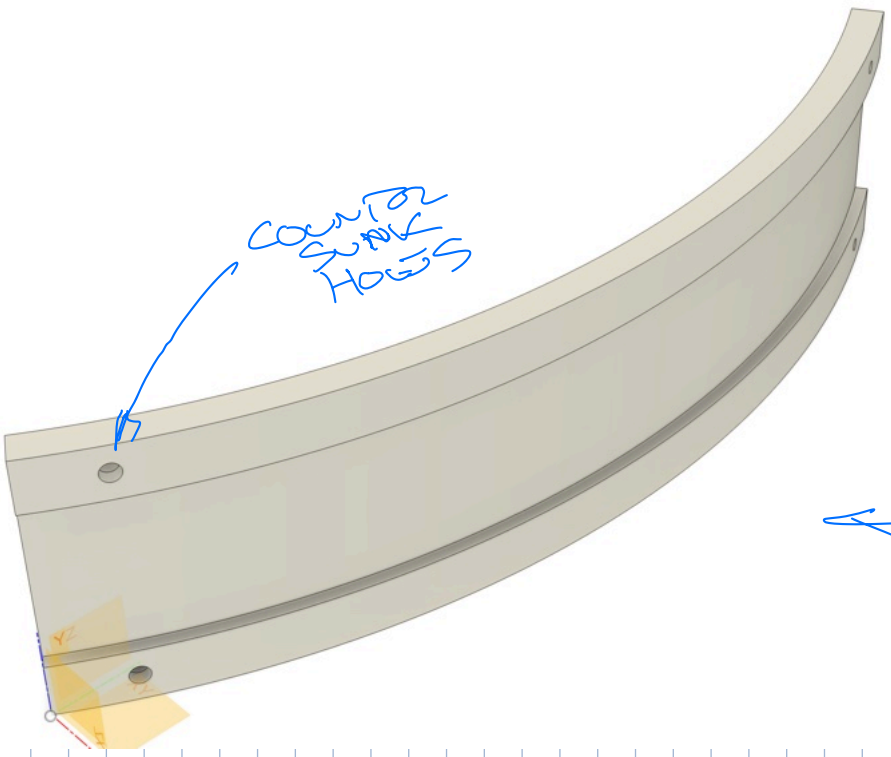
3D PRINTING MAIN BOX TODAY

OK! BOX DESIGN IN FUSION 360 LOOKS PRETTY GOOD



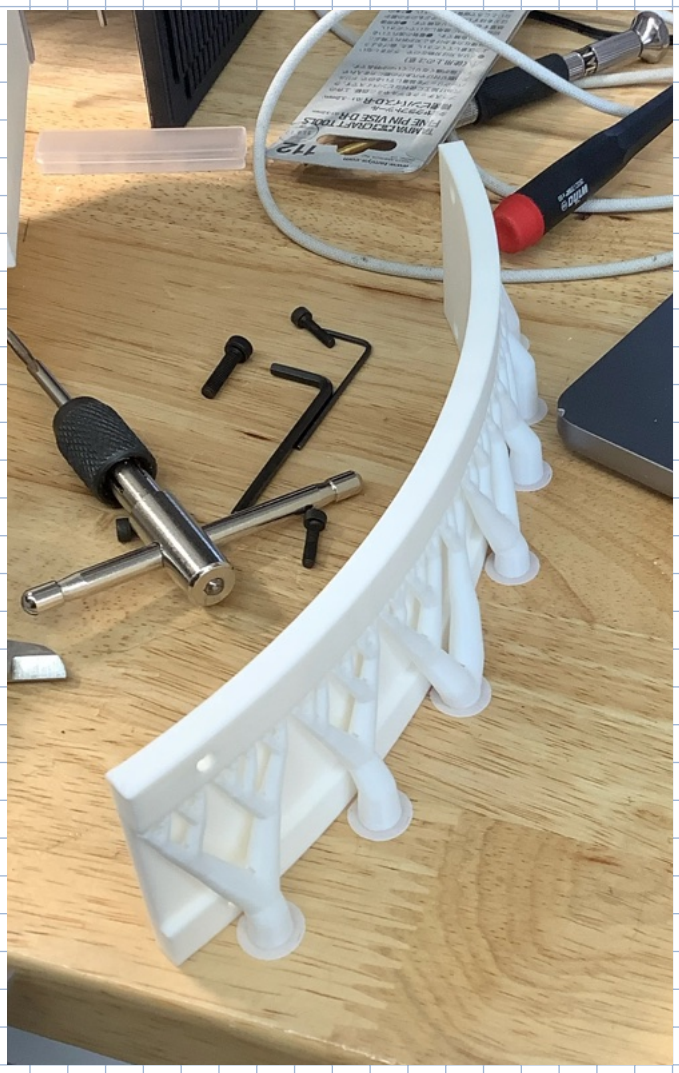
SLOTS TO HOLD ELECTRONICS BOARD





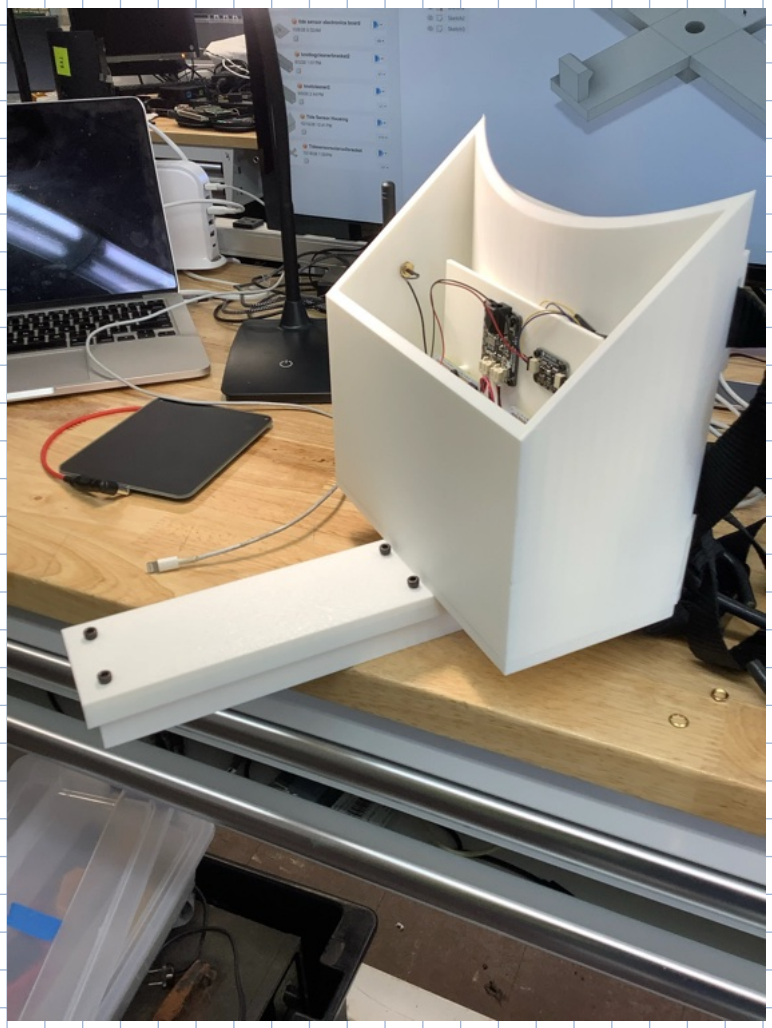
COUNTER SUNK HOLES

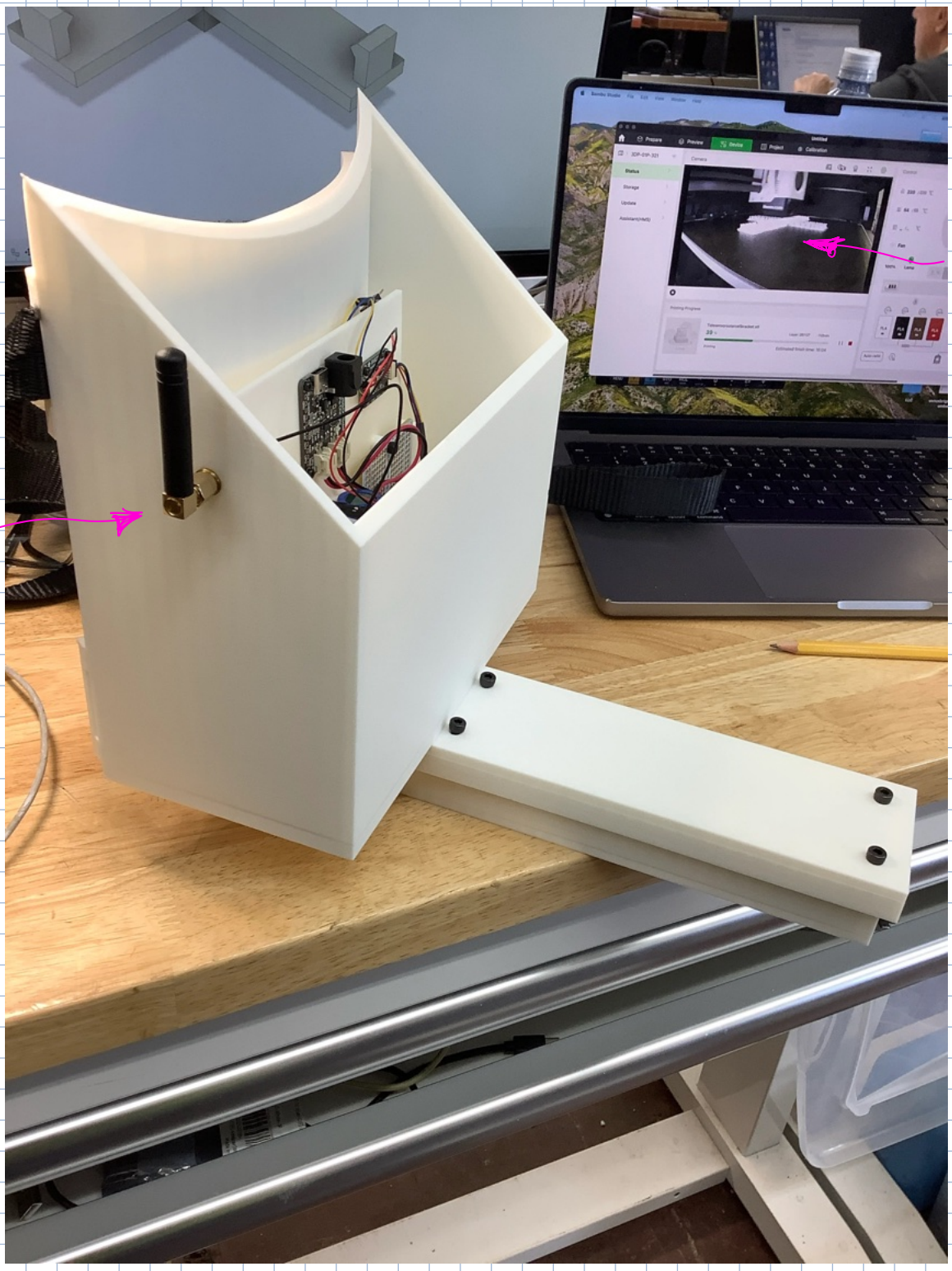
SNAP HOLDER (2)









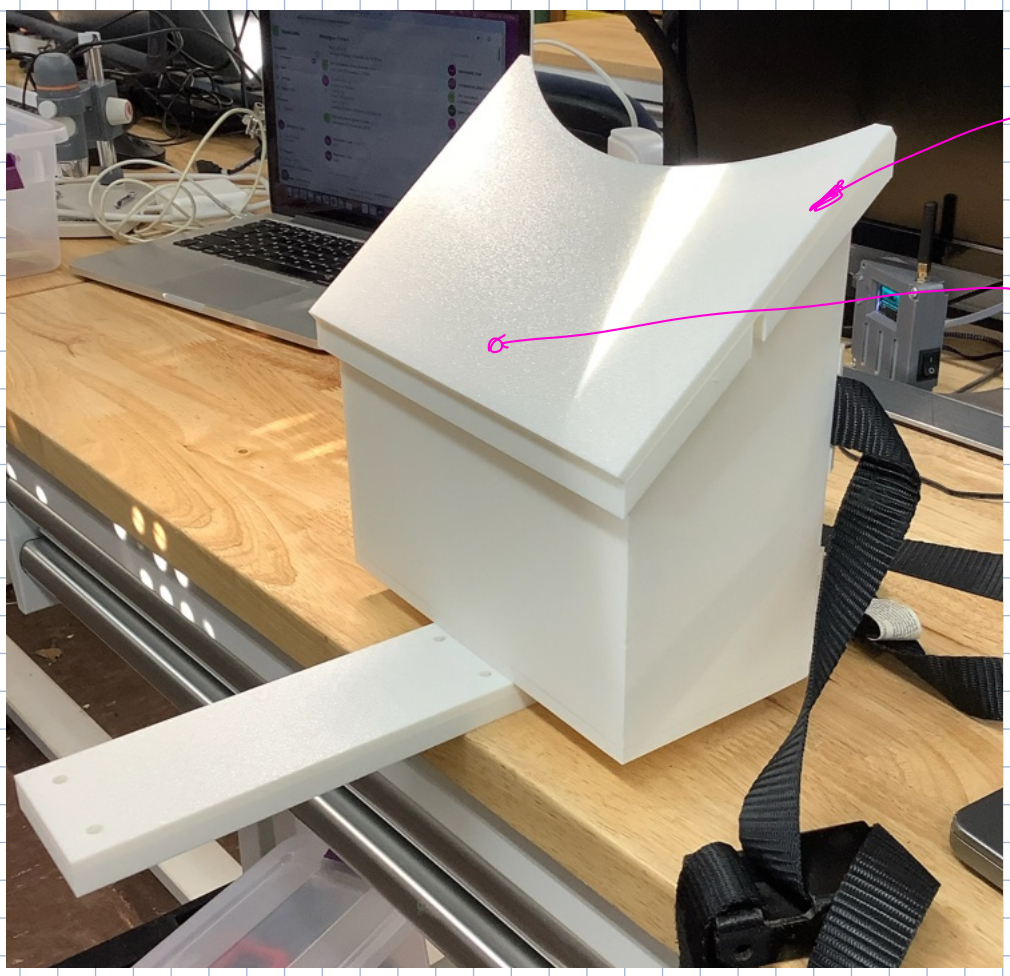

SOLAR CELL BRACKET BEING PRINTED
DRILLED 1/4" HOLE FOR LORA ANTENNA

10/20/2026 (TUESDAY)

PRINTED TOP - MADE CHANGE

INSTALLED FINAL SOFTWARE (WITH MODIFICATIONS)

SET UP SOLAR PANEL BRACKET





TOP MOVED STRAP BRACKET (TOO CLOSE TO TOP)

TAPPED NEW LOWER HOLES, BUT SCREWS ARE NOT HELD WELL - NEED TO GLUE

MADE NOTCH IN CAD TO ACCOMODATE SOLAR PANEL WIRE

TO DO:
CUT NOTCH IN MAIN BOX FOR SOLAR PANEL WIRE

* NEED TO GET RTV OR OTHER SEALANT TOO

NEW SOFTWARE IS WORKING, BUT ONLY IF I SEND ONLY AVG & SD OF DATA. RAW DATA APPEARS TO BE TOO MUCH (WHY?)
└ 10 POINTS

CAD FIXES: (FOR FUTURE PRINT)

- SOLAR CELL BRACKET M4 BOLT HOLE IS TOO WIDE
- MOVE SCREW HOLE POSITIONS ON MAIN BOX LOWER FOR UPPER STRAP BRACKET TO MAKE ROOM FOR TOP
- MAKE NOTCH IN MAIN BOX FOR SOLAR CELL WIRE
- TAKE TOP BACK TO 100% (FROM 103% (?)) SO EVERYTHING LOOKS GOOD IN CAD AND THEN SCALE IT UP IN BAMBU SOFTWARE BEFORE PRINTING
- CONSIDER SCALING UP HOLES IN MAIN BOX AND EXTENSION ARM TOP TO ACCOMODATE M3 SCREWS INSTEAD OF M2 FOR MORE STRENGTH

TTD BEFORE DEPLOYING

- GET GASKET COMPOUND & APPLY
- CUT STRAPS TO APPROPRIATE LENGTHS
- GLUE & SCREW
- MAKE SCREW HOLES TO SECURE TOP TO MAIN BOX
- CHARGE UP BATTERY FULLY
- SECURE SOLAR PANEL
- TEST!! BUT WHERE?



WORKS WITH PAIL OF WATER. (MOSTLY)

10/23/26

TIDE SENSOR IS STILL IN PLACE ON MEF, BUT SOMEONE UNPLUGGED THE LAPTOP SO I DON'T KNOW IF IT RAN CONTINUOUSLY!! ONLY DATA FOR ~ 4 HOURS WAS RECORDED. - WHY SO LITTLE?

I SET UP MONITORING ON THE CDL LOGIN ON MY OLD LAPTOP (INSTEAD OF MY PERSONAL ONE)

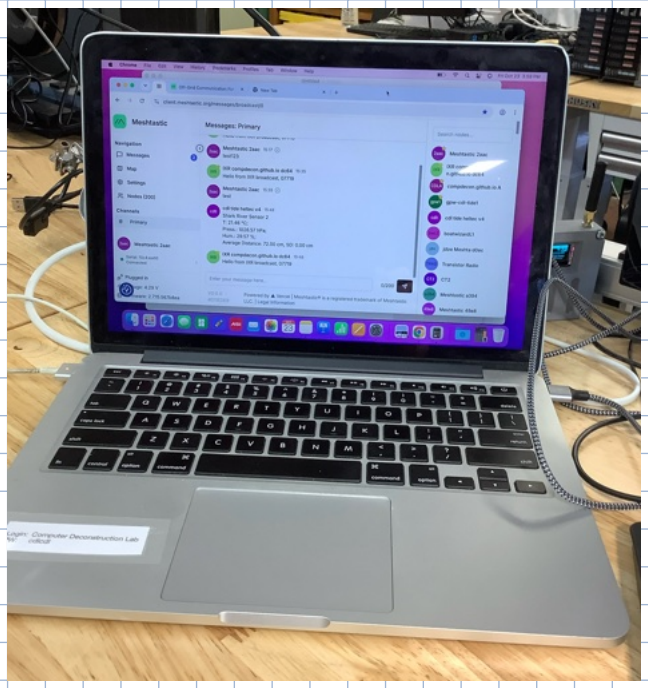
SYSTEM SETTINGS: NEVER SLEEP

OK! GOT SIGNAL @ 15:48 10/23/26

72 CM READING TO BOARD (0 CM S.D.)

ACTUAL MEASUREMENT: 28.5 IN = 72.4 CM!

OK!



OK!
GOT ANOTHER READING @ 16:04
(70.0 CM 5.0 CM S.D.)
SO SENSOR SEEMS TO BE RUNNING

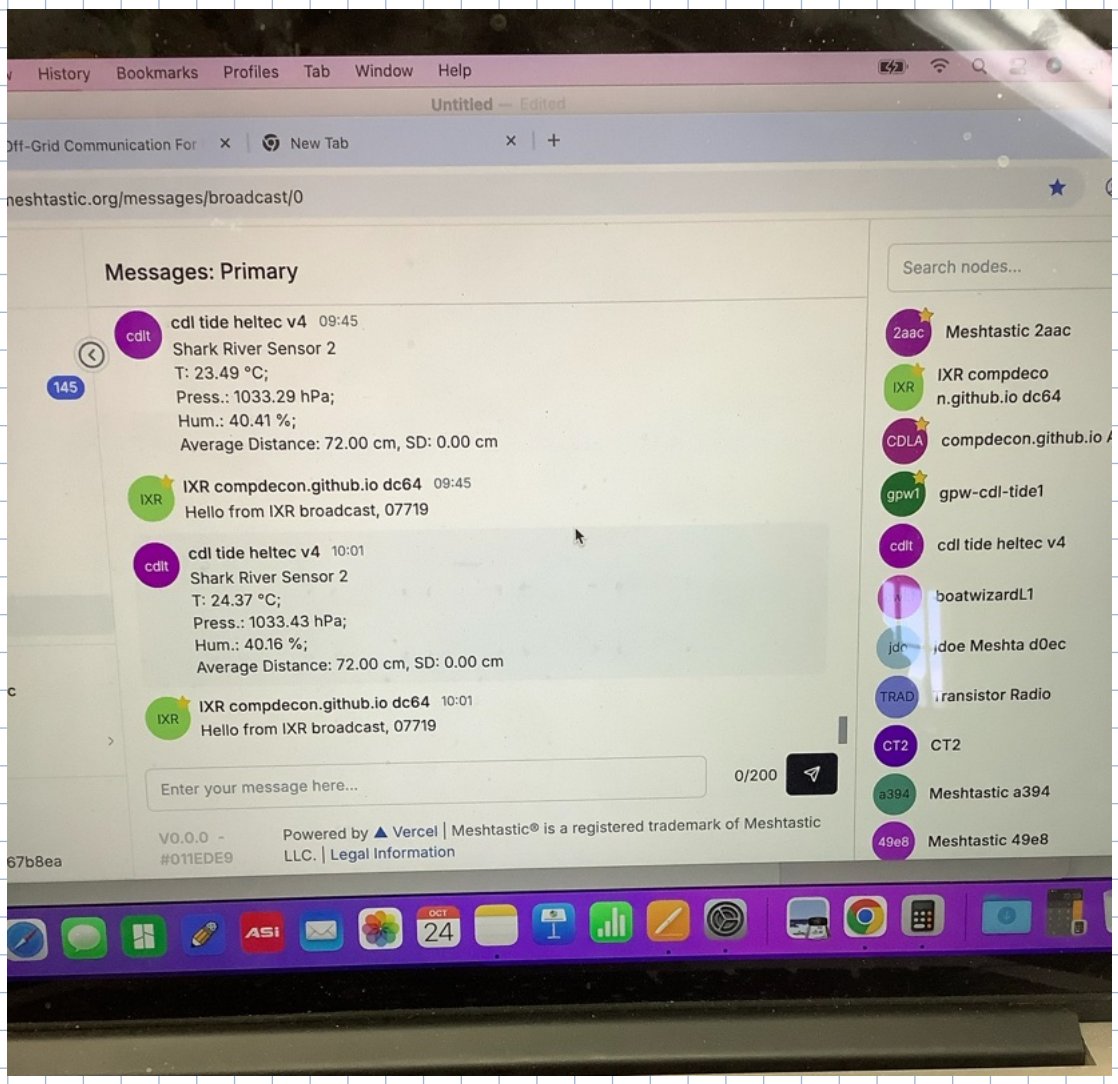
BOUGHT GASKET SEAL, SILICONE SEAL, AND CLEAR UV PROTECTION SPRAY COAT FROM TAYLORS TO WATERPROOF UNIT IF THINGS LOOK GOOD WITH OVERNIGHT READINGS



IS THE REASON THAT NOT ALL OF THE RAW DATA IS NOT RECEIVED IS BECAUSE THE IXR MESHTASTIC DEVICE SPITS OUT ITS CANNED MESSAGE BEFORE TIDE SENSOR IS FINISHED - LOSING THE LAST MESSAGE LINES??

SEND ALL DATA IN ONE TEXT
(1) (SINGLE PRINTLN)
OR
(2) STOP THE CANNED MESSAGE RESPONSE

SATURDAY, OCTOBER 24, 2026
OK! SENSOR RAN THROUGH THE NIGHT!



READINGS EVERY 15 MINUTES FROM 4 PM YESTERDAY TO 10 AM THIS MORNING

NO MISSING DATA

28 ± 1/2" → 72.4 ± 1 cm.
MEASURED WITH OIL STICK

OVER 1" RIGHT ON TARGET

MOVED BOARD TO 36.0" (91.4 cm)

READING: 88 ± 12 cm

(BIG SPREAD)

UPDATE 10/26/26
TOOK SENSOR OFF TREE
GLUED, SILICONED, & UV CLEAR-COATED HOUSING

10/29/26 WOW! SENSOR STILL WORKING FROM MONDAY AFTERNOON TO WEDNESDAY AFTERNOON WITHOUT SOLAR RECHARGING!!

↳ ∴ MORE THAN ENOUGH BATTERY POWER TO ENDURE CLOUDY WINTER DAYS

INSTALLED NEW SOFTWARE THAT INCLUDES ALL OF THE RAW DISTANCE DATA

WORKS. — THEORY THAT IXR MESHTASTIC "HELLO" BROADCAST INTERFERES WITH SENDING DATA SEEMS TO BE TRUE. BY SENDING ALL DATA AT ONCE, NOTHING GETS LOST

READY TO DEPLOY!

MATT WILL HELP @ 11 AM FRIDAY